

ZAINAB KASHIF

Software Engineer

Next.js • Performance Optimization • Design Systems • IEEE Published Researcher
Pakistan

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Education

National University of Sciences and Technology

Bachelor of Science in Computer Science

Sep. 2021 – May 2025

Islamabad, Pakistan

Experience

Carbonteq

Software Engineer

May 2025 – Present

Pakistan

- Develop and maintain scalable frontend applications using Next.js and modern JavaScript frameworks, following clean architecture and industry-standard conventions.
- Improved application performance and accessibility, achieving significant gains in Lighthouse scores through optimization techniques such as code-splitting, lazy loading, and efficient rendering strategies.
- Contributed to and extended in-house design systems while integrating external design systems, ensuring UI consistency, reusability, and accessibility across products.
- Collaborated closely with designers and backend engineers to deliver seamless, production-grade features with strong focus on user experience.
- Built and integrated RESTful APIs and occasionally contributed to backend services, improving end-to-end system reliability.
- Maintained high code quality through reusable component design, structured codebases, and active participation in code reviews.

NiftyBooks

AI Project Lead (Machine Learning Engineer)

July 2024 – May 2025

Germany, Remote

- Led end-to-end development of AI-powered illustration tools, managing timelines, deliverables, and cross-functional collaboration in a distributed team.
- Owned sprint planning and task prioritization using Agile methodologies, ensuring alignment between engineering, design, and business stakeholders.
- Collaborated with founders and clients to translate business requirements into scalable AI solutions.
- Directed development and evaluation of SDXL and FLUX LoRAs for stylized image generation in production environments.
- Delivered technical documentation, demos, and stakeholder updates using G-Suite and Confluence.

Innonexus Tech

Machine Learning Intern

May 2023 – August 2023

India, Remote

- Developed a legal advisory chatbot using LangChain and NLP pipelines, improving contextual accuracy of legal information retrieval.
- Built full-stack components integrating responsive frontend interfaces with backend APIs.
- Optimized NLP pipelines, reducing processing time by 35% and improving entity recognition accuracy.
- Implemented retrieval-augmented generation (RAG) for context-aware legal recommendations.

Projects

Custom Point of Sale (POS) System | *Next.js, Square POS, REST APIs*

2025

- Designed and developed a full-featured POS system integrated with Square POS, enabling real-time transactions for retail workflows.
- Implemented robust pricing logic handling dynamic taxes, discounts, and edge cases to ensure accurate and reliable transaction processing.
- Architected a scalable frontend using Next.js, following clean code principles, modular component design, and maintainable folder structures.
- Designed the complete UI/UX and established a reusable design system to ensure consistency, accessibility, and rapid feature development.
- Integrated backend APIs for order and payment workflows, delivering a seamless end-to-end checkout experience.

Style Change Detection System | *PyTorch, LoRA, Mistral-7B*

November 2024

- Developed a model to detect stylistic shifts in text, targeting applications in authorship analysis and content integrity.
- Fine-tuned Mistral-7B using LoRA for efficient training, reducing computational overhead while maintaining strong performance.
- Engineered preprocessing pipelines, tokenization strategies, and evaluation metrics including F1-score and confusion matrices.

Music Generation based on Prompt | *TensorFlow, Transformers*

January 2024

- Built a prompt-driven music generation system that adapts output based on user mood and emotional context.
- Integrated an emotion classification model with an LSTM-based generator, achieving 80% mood prediction accuracy.

AI-Based Car Speed Detection System | *YOLOv8, DeepSORT*

February 2024

- Developed a real-time vehicle monitoring system for speed estimation in traffic analysis scenarios.
- Combined object detection, tracking, and motion analysis techniques to achieve 95% accuracy in speed estimation.

Publications

- **Zainab Kashif et al.**, “Multi-Modular Diagnosis and Stage Classification of Alzheimer’s Disease,” *IEEE Access*, 2025. Proposed a multi-modal deep learning framework combining MRI imaging and clinical data for improved early-stage Alzheimer’s detection. [View Publication]

Technical Skills

- Languages:** TypeScript, Python, SQL, C/C++
Frontend: Next.js, React.js, Design Systems, Responsive Design, State Management
Backend: Node.js, Express.js, REST APIs, Flask
Tools: Git, GitHub, VS Code, Jupyter Notebook
Technologies: TensorFlow, PyTorch, scikit-learn, Pandas, NumPy, Matplotlib
Concepts: Performance Optimization, Accessibility, Clean Architecture, Agile, SDLC
Research: Experimental Design, Model Evaluation, LaTeX, Scientific Writing

Certifications

Machine Learning Specialization

Nov 2023 – Jan 2024

DeepLearning.AI
Coursera

- Supervised, Unsupervised Learning, and Advanced ML Algorithms

Web Development

Aug 2023 – Sep 2023

Meta
Coursera

- React, JavaScript, Version Control

Computer Networking

Dec 2023

Cisco
Cisco Networking Academy

- Networking Essentials